

Saillesh Somasundaram

+1 (647)-619-7877 | s5somasu@uwaterloo.ca | [linkedin/sailshsomasundaram](https://www.linkedin.com/in/sailshsomasundaram)

TECHNICAL SKILLS

Languages: Python, SQL, Bash, PowerShell, JavaScript, Scala, Ruby, Java

Tool/Infrastructure: Linux (Ubuntu/RHEL), GitLab CI/CD, Jenkins, Docker, Kubernetes, Terraform, Ansible, AWS(EC2, S3, Lambda, ECS, CloudWatch), Azure (AKS), Prometheus, Grafana, GitOps, ELK Stack (Elasticsearch, Logstash, Kibana)

EDUCATION

University of Waterloo

September 2021 – April 2026

Bachelor of Science in Honors Physics

Waterloo, ON

- **Technical Proficiency:** Signals & Systems, Data Structures, Algorithms, Computational Physics, Computational Modeling
- **Recipient of:** Presidential Scholarship and Term Distinction

EXPERIENCE

DevOps Engineer, Symcor Inc

September 2025 - Present

Internship

Toronto, ON

- Hardened CI/CD pipelines by enforcing quality gates + secret scanning at merge time, integrating SonarQube analysis reports across 100+ repositories
- Engineered a GitLab Runner health monitoring system using API telemetry, cutting CI brownouts and stalled jobs by 40%
- Modernized Java CI workflows by upgrading Maven and integrating JaCoCo coverage instrumentation directly into pipelines
- Designed a Bash-based multi repository scanner to traverse GitLab groups and detect deprecated API usage patterns
- Extended Ansible automation with reusable roles managing service provisioning, secrets injection, and environment parity across dev, staging, and production hosts

Quality Automation Engineer, OmniaBio

August 2024 – December 2024

Internship

Hamilton, ON

- Engineered Python-based anomaly detection system analyzing 8K+ pharmaceutical samples using statistical process control (Z-scores, control charts), reducing manual QC review cycles by 60%
- Automated data validation pipelines with schema enforcement and outlier detection, cutting GMP inspection turnaround time by 35% while maintaining 99.8% compliance rate

Data Analyst, PepsiCo Inc

January 2023 – April 2024

Internship

Mississauga, ON

- Designed optimized SQL aggregation layers for Power BI dashboards, reducing query latency and improving KPI drill down
- Applied regression and time-series modeling to operational datasets, identifying 3 bottleneck processes causing 25% labor inefficiency and enabling \$180K in annualized labor cost savings
- Automated mission-critical reporting workflows with SLA enforcement (99.9% on-time delivery), supporting 500K+ internal users across 15 markets and eliminating 20 hours/week of manual data reconciliation

Software Engineer, UWAFt

September 2022 - January 2023

Design Team

Waterloo, ON

- Developed a React-based VR simulation front end to support autonomous vehicle testing and validation workflows
- Designed Ansible automation managing inventories, host groupings, and service mappings across environments
- Instrumented pipeline reliability metrics and surfaced failure patterns using Prometheus and Grafana dashboards

PROJECTS

ICT Displacement and Liquidity Sweep Detector *TradeView • Pine Script*

- Implemented a stateful streaming signal engine using custom data structures to track sweep events, and displacement confirmation
- Designed deterministic long/short trade setups with configurable sensitivity for systematic backtesting

Heston Stochastic Volatility Model Engine *Python • Google Vision • APIs*

- Architected production-grade stochastic volatility engine implementing Heston model with three independent solvers: ADI finite differences, Fourier inversion, and Monte Carlo (QE scheme)

Real time Stock News Sentiment *Python • REST APIs*

- Built event-driven NLP pipeline ingesting 10K+ news articles/day from FinHub, Alpha Vantage filings using REST APIs and asyncio for concurrent processing